MIL-E-1/923C
25 August 1966
SUPERSEDING
MIL-E-1/923B
18 December 1963
(See note 13)

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 6460

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Reflex, integral cavity, tunable, waveguide output

ABSOLUTE RATINGS: See note 1

Parameter:	Ef	Ers	Irs	Prs	Er	Ehk	Body temperature
Unit:	V	Vdc	mAdc	W	Vdc	Vdc	° C
Maximum:	7.0	775	80	60	-1,000	±45	160
Minimum:	5.7	---	---	---	(see note 2)	---	--- (see note 4)

PHYSICAL CHARACTERISTICS:

See figure 1
Cathode: Unipotential
Base: B6-242

TEST CONDITIONS:

Parameter:	Ef	Ers	Er	Ehk	Body temperature
Unit:	V	Vdc	Vdc	Vdc	° C
1. (see notes 3 and 4)	6.3	750	Adjust	0	100 (max)
2. (see notes 3 and 4)	6.3	750	-250 to -400	0	100 (max)
3. (see note 4)	6.3	300	Adjust	0	100 (max)

REQUIREMENT OR TEST:

GENERAL

3.2 Qualification - Required

5. Preparation for delivery

E-50.2 Holding period (t = 168 hours)

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Qualification					
Appendix D, 20(b)	Dimensions	See figure 1	---	---	---	---
1023	Temperature coefficient	Test condition 2; Er/max Po; TA = 0° C to + 80° C; F = 6.12 ±0.3% Gc (see note 7)	ΔF/ΔTA	---	±70	kc/° C
---	Effective resonator position	F = 6.27 ±0.3% Gc (see note 8)	Length	0.797	0.847	in.
	Quality conformance inspection, part 1					
---	† Emission	Test condition 1; Ef = 5.7 V (see notes 9 and 10)	ΔIrs/Irs	---	-15	%
4212	Resonator current	Test condition 1	Irs	---	80	mAdc
4213	Reflector voltage (1)	Power output (1)	Er	-350	-445	Vdc
4213	Reflector voltage (3)	Power output (3)	Er	-95	-175	Vdc
4229	† Total reflector current	See notes 10 and 11	Ir	---	5	μAdc
1301	† Heater current	Ef = 6.3 V	If	0.7	0.9	A
---	Tunable frequency	Test condition 1; Er/max Po (see note 12)	F	5.30	6.60	Gc
4211	Power output (1)	Test condition (1); Er/max Po; F = 6.60 ±0.3% Gc	Po	0.7	---	W
4211	Power output (3)	Test condition (3); Er/max Po; F = 5.30 ±0.3% Gc	Po	25	---	mW
4201	Electronic tuning range (1)	Test condition (1); Er/50% max Po (see note 5)	ΔF	28	---	Mc
4201	Electronic tuning range (3)	Test condition (3); Er/50% max Po; F = 5.30 ±0.3% Gc	ΔF	15	---	Mc
	Quality conformance inspection, part 2					
Appendix D, 20(b)	Dimensions	See figure 1	---	---	---	---
1211	Insulation of electrodes	Ehk = ±45 Vdc	Ihk	---	250	μAdc
4211	Power output (2)	Test condition (2); Er/max Po; F = 6.12 ±0.3% and 6.42 ±0.3% Gc	Po	0.7	---	W

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
	Quality conformance inspection, part 2 -Continued					
4201	Electronic tuning range (2)	Test condition (2); Er/50% max Po (see note 5)	ΔF	32	---	Mc
---	Reflector voltage (2), modulation coefficient	Test condition (2); power output (2); ΔF = ± 2 Mc	ΔF/ΔEr	275	500	kc/V
	Quality conformance inspection, part 3					
4.7	Life-test provisions	Group D; power output (2); TA = + 40° to + 50° C (see note 6)	t	2,000	---	hr
4.7.3	Life-test end point	Power output (2)	Po	0.5	---	W

NOTES:

1. Ratings should not be exceeded under continuous or transient conditions. A single rating may be the limitation and simultaneous operation at another rating may not be possible. Design values for systems should include a safety factor to maintain operation within ratings under voltage and environmental variations. The life warranty is predicated on operation of the tube under the specified test conditions.
2. The reflector voltage must always be negative with respect to the cathode by at least 20 volts.
3. At least 30 cfm of air cooling shall be used during testing.
4. All oscillation tests shall be made with the tube rigidly connected to a UG-344/U flange, as specified in MIL-F-3922/6, on appropriate RG-50/U waveguide equipment, in accordance with MIL-W-85. The load VSWR for the tube shall be less than 1.1:1. For maximum tube life, the tube temperature should be less than 100° C.
5. Electronic tuning range is measured between the half-power (3 db) points of the reflector voltage mode.
6. All electrode voltages shall be maintained constant during the test.
7. Constant air cooling must be used during the test.
8. The tube under test shall be adjusted to its center (resonance) frequency. Tune the signal source - 150 Mc from the tube's center frequency and locate the position of the null on a standing wave indicator. Repeat the procedure with the signal source tuned to + 150 Mc from the tube's center frequency. Replace the tube under test with a shorting plate on the waveguide; tune the signal source to the tube's center frequency and locate the position of the null on the standing wave indicator. The effective resonator position is then defined as the position of the VSWR null with the shorting plate attached to the waveguide minus the average location of the VSWR null at ± 150 Mc from the tube's center frequency with the tube attached to the waveguide. The tube shall not be operating during the test.
9. The resonator current shall not drop more than the specified percentage when the heater voltage is decreased from the normal operating potential to the lower specified value.

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NOTES: -Continued

10. The tube shall not be oscillating during the test.

11. After 1 minute with all voltages applied, total reflector current shall not exceed the specified limit.

12. The tube shall be capable of tuning over the specified range of frequencies with a single change of reflector voltage mode.

13. CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDES THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Custodians:
Army - EL
Navy - SH
Air Force - 11

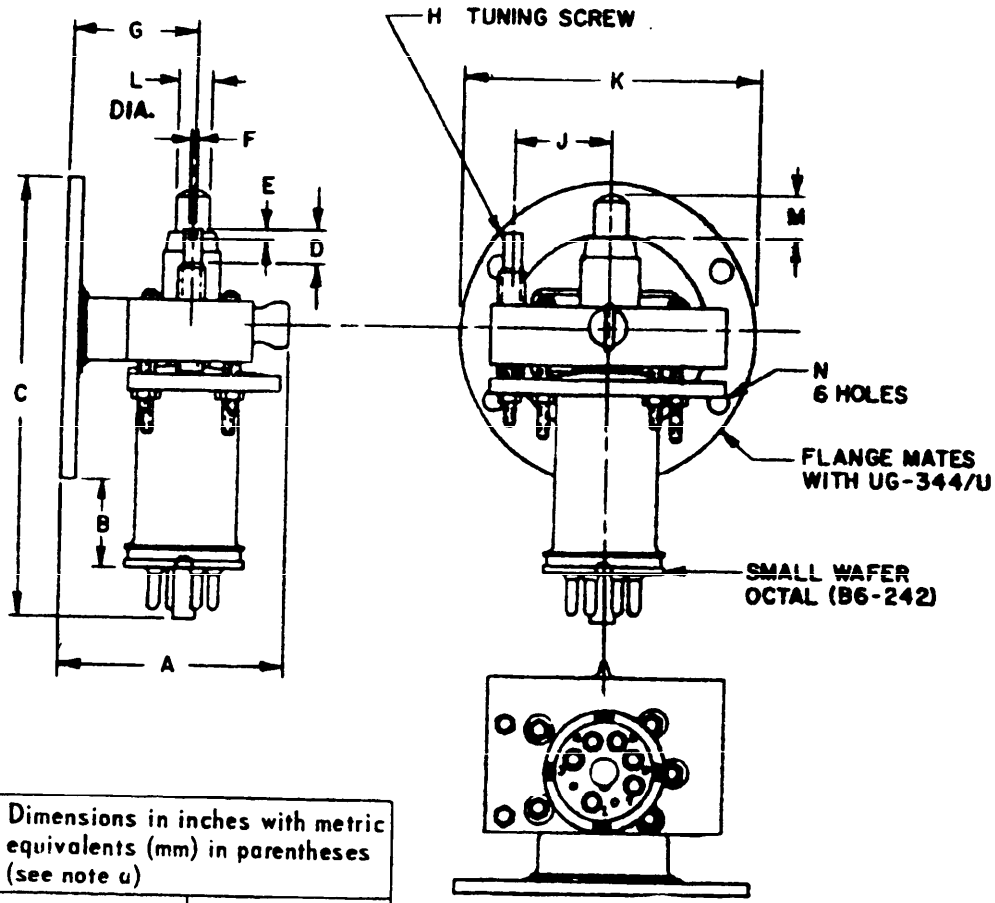
Review activities:
Army - EL
Navy - SH, EC
Air Force - 11, 17, 85

User activities:
Army - MU
Navy - AS, OS, MC, CG
Air Force - 14, 19

Preparing activity:
Navy - SH

(Project 5960-2215-14)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note a)	
	Minimum	Maximum
Qualification		
B	.935 (23.75)	.945 (24.00)
D	.370 (9.40)	.380 (9.65)
E	.090 (2.29)	.100 (2.54)
F	.060 (1.52)	.070 (1.78)
H	.245 (6.22)	.250 (6.35)
N	.215 (5.46)	.225 (5.72)
Quality conformance inspection, part 2		
A		2.625 (66.68)
C		4.750 (120.65)
G	1.280 (32.51)	1.340 (34.04)
J	1.040 (26.41)	1.090 (27.43)
K		3.190 (80.77)
L	3.55 (90.17)	3.65 (92.71)
M		.426 (10.82)

PIN	CONNECTIONS
1	SHELL
3	NO CONNECTIONS
4	CATHODE
5	HEATER
6	HEATER
7	INTERNALLY CONNECTED
CAP	REFLECTOR

NOTES:

a. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Outline drawing of electron tube type 6460.

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